

Alginate-kaolinite beads incorporated with agrochemicals: new baits for leaf-cutting ants control

Valber G. O. Duarte

Universidade Federal de Viçosa - Campus Rio Paranaíba

Victor Leite

Universidade Federal de Viçosa - Campus Rio Paranaíba

Danúbia A. C. Nobre

Universidade Federal dos Vales do Jequitinhonha e Mucuri

Renê C. Silva

Universidade Federal de Viçosa

Vera R.L. Constantino

Universidade de São Paulo

Gustavo F. Castro

Universidade Federal de Viçosa

Flávio L. Fernandes

Universidade Federal de Viçosa - Campus Rio Paranaíba

Jairo Tronto

jairotronto@ufv.br

Universidade Federal de Viçosa - Campus Rio Paranaíba

Research Article

Keywords: pest control, insecticide, baits, beads, *Atta sexdens*

Posted Date: May 7th, 2024

DOI: <https://doi.org/10.21203/rs.3.rs-4331320/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

Insect pests such as leaf-cutting ants (genera *Atta* spp. and *Acromyrmex* spp.) are a major problem in tropical countries because they can cause significant damage to crops, resulting in substantial production losses. These insects have been controlled chemically using granular baits. Although these granules are cheap, they can pose some problems due to their low resistance to humidity. Therefore, developing new formulations that present environmental resistance, sustained release of the active chemical, and good ant-killing efficacy can expand the management options. This work aimed to produce new baits in bead form to control leaf-cutting ants (*Atta sexdens*). The beads are derived from the interaction between organic and inorganic materials, synthesized from mixtures of alginate polymer, kaolinite clay, sodium tetraborate, *Beauveria bassiana* spores, and chlorpyrifos. X-ray Diffraction, Fourier Transform Infrared Spectroscopy, Thermogravimetric Analysis Coupled with Differential Scanning Calorimetry and Mass Spectrometry, and Scanning Electron Microscopy provided important information about the structure and composition of the produced baits. For all baits, the data of the "*in vitro*" release study of boron were adjusted according to Higuchi's kinetic model. The results of boron leaching in soil columns experiments showed that for some baits, 100% of B leached after 15 days. According to the laboratory and field bioassays results, the beads showed good attractiveness and efficiency in controlling leaf-cutting ants. The synthesis of new beads of alginate-kaolinite containing agrochemicals proved to be a suitable bait with ant-killing potential and efficient for the chemical control of *A. sexdens*.

Introduction

Leaf-cutting ants of the genera *Atta* spp. and *Acromyrmex* spp. (Hymenoptera: Formicidae) are one of the most severe insect pests in the world (Moreira et al. 2004; Pérezet al. 2011; Furtado et al. 2013; Boulogne et al. 2014). These insects reduce the vegetal mass of various crops at all stages of plant development, resulting in significant production losses. Leaf-cutting ants can reduce plant biomass by up to 15% and cause mortality in young plants (Urbas et al. 2010; Swanson et al. 2019). Losses caused by these species reach approximately \$8.26 per eucalyptus tree, and the ant control costs approach 30% of the production costs (Della Lucia 2011). Among the ant species found in Brazil, *Atta sexdens* L. (Hymenoptera: Formicidae) is one of the most aggressive, with a wide geographic distribution, polyphagous behavior, and difficult to control (Teixeira et al. 2023).

There are several ways to deal with leaf-cutting ants, including mechanical, cultural, biological, and chemical control techniques (Zanetti et al. 2003a; Reis et al. 2015). The most efficient and widely distributed method is the chemical one (Della Lucia 2011). One popular approach is to use ant baits, which are inexpensive and easy to apply locally. This method has been proven highly efficient in the field without requiring expensive equipment (Cruz et al. 1996; Zanetti et al. 2003b).

There are many active products currently used to control leaf-cutting ants. In agriculture are sulfluramid ($C_{10}H_6F_{17}NO_2S$), chlorpyrifos ($C_9H_{11}Cl_3NO_3PS$), and fipronil ($C_{12}H_4Cl_2F_6N_4OS$) (MAPA 2003). Chlorpyrifos is probably the most widely used insecticide in agriculture due to its low cost, easy

application, and broad spectrum of action (Lima et al. 2011). This insecticide acts through contact, ingestion, and fumigation (Nakagawa et al. 2016; Wołejko et al. 2022). However, due to their high toxicity and persistence in the environment, they can lead to the contamination of the ecosystems. One biological agent that shows toxicity to ants is the *Beauveria bassiana* fungus, which is commercially available and widely used in agriculture and forestry (Li et al. 2011). It has been extensively studied, showing prospects for future success in ant management (Cantú-Ruiz et al. 2017; Folgarait et al. 2020; Loureiro et al. 2022). The success of using *B. bassiana* may be true because it has been observed that there are isolates of *B. bassiana* that cause mortality of 87% of *A. sexdens* workers after 4 days of application (Loureiro et al. 2005). On the other hand, some ant control products on the market contain sodium tetraborate ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$). These products promise to repel and eliminate insects, especially ants, by altering their digestive system and overall behavior by acting as a disruptor, preventing ants from functioning normally and eventually leading to their death. Studies have reported toxicity and repellency to the Argentine ant - *Linepithema humile* Mayr and the *Anoplolepis tenella* Santschi (Hymenoptera: Formicidae) ant exposed to compounds containing boron (B), such as boric acid, disodium octaborate tetrahydrate, and borax (Hooper-Bui et al. 2000; Klotz et al. 2000; Sola et al. 2013; Hanna et al. 2015; Suiter et al. 2021). Hence, using materials that can act as matrices for the gradual release of active leaf-cutting ant control products can be an effective alternative to reducing the use of traditional chemical compounds for insect pest management. A promising approach to ant control is to formulate chlorpyrifos, *B. bassiana* spores, and sodium tetraborate within polymeric organic-inorganic hybrid matrices in bead form. This approach differs from conventional reliance on chemical compounds and can be a more sustainable method for managing ant populations (Vieira et al. 2014).

Notably, in Brazil, sulfluramid (0.2%) baits are the most commonly used to control leaf-cutting ants, as they show good results on several species. These baits come in the form of pellets containing the active ingredient sulfluramid and dehydrated orange pulp dissolved in soybean oil (Forti et al. 2007; Buratto et al. 2017). However, these commercial materials have low humidity resistance, which can cause the degradation of the bait before contacting the insect, resulting in low control efficacy (Peregrine et al. 1976; Zanúncio et al. 1992). Therefore, there is a need to produce more efficient baits with better environmental resistance, sustained release of the active chemical, and good ant-killing efficacy.

In the context of controlling leaf-cutter ants, the use of organic-inorganic hybrids, shows promise in improving the effectiveness of control methods and promoting environmental sustainability (Fernandes et al. 2014). These materials are expected to have promising properties due to the synergistic effect between their components, which are not found in the original materials (Sanchez et al. 2011a; Sanchez et al. 2011b; Abellán et al. 2015). To form improved materials, renewable biopolymers and 2D inorganic materials, such as clays, layered double hydroxides, and graphene, are combined to form organic-inorganic hybrid materials (Darder et al. 2005; Sanchez et al. 2011a). These materials are being applied in various fields, including pharmaceuticals (Takahashi and Yamaguchi 1991), agriculture (Benício et al. 2015), environment (Sanchez et al. 2011b), chemistry (Sanchez et al. 2011a) and others.

Sodium alginate ($\text{NaC}_6\text{H}_7\text{O}_6$) is a polysaccharide obtained from marine algae, a derivative of alginic acid formed by 1,4- β -D-mannuronic (M) and α -L-guluronic (G) acids. This polymer has interesting properties such as biodegradability, non-immunogenicity, and the ability to form gels with various crosslinking agents under mild conditions (Cunha et al. 2010; Chan et al. 2016). In aqueous solutions, when the sodium alginate comes into contact with divalent metal cations, such as Ca^{2+} , Mg^{2+} , or Ba^{2+} , due to the presence of carboxyl groups (COO^-) in its structure, a hydrophobic gel is formed by crosslinking (Haug et al. 1970; Jain and Datta 2016). Thus, this polymer represents a potential carrier for the sustained delivery of chemical actives (Wan et al. 1994).

Inorganic modifiers include carbon nanotubes, active carbon, natural clays, and activated clays. The activated clays have been widely used as fillers due to their abundance and the possibility of structural modification of the layers. Kaolinite is a clay mineral ($\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$) with a chemical composition of 39.5% of Al_2O_3 , 46.5% of SiO_2 , and 14.0% of H_2O . This clay has excellent absorption and stabilization properties that promote the sustained release of active chemical ingredients when they are part of the organic-inorganic hybrid material (Etchepare et al. 2015).

The scientific literature has not yet reported the synthesis of organic-inorganic hybrid materials made from alginate-kaolinite beads containing sodium tetraborate, *B. bassiana* spores, and chlorpyrifos as baits for leaf-cutting ant control. However, the physicochemical interaction between the pristine parts of the organic-inorganic hybrid material can modify the stability of the baits under environmental conditions and the characteristics of reactivity, dissolution, and the release of the active chemical ingredient.

The present work aimed to synthesize and characterize ant baits prepared from alginate-kaolinite beads containing sodium tetraborate, *B. bassiana* spores, and chlorpyrifos; evaluate the "*in vitro*" and leaching release of boron from the produced baits, and compare the results with a commercial source; and evaluate the effectiveness of the produced baits in the control of *A. sexdens* through bioassays under field and laboratory conditions.

Materials and Methods

Chemicals

All H_2O used was previously distilled and deionized using the Millipore MilliQ[®] system. The following reagents were used in the synthesis of the baits: sodium chlorite - CaCl_2 (Dynamic, 99%), sodium alginate - $\text{NaC}_6\text{H}_7\text{O}_6$ (Exodus, 90.8–106%), sodium tetraborate - $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ (Reagen, 99–100%), *B. bassiana* (Boveril[®], 2.0×10^9 conidia g^{-1} , WP), chlorpyrifos - $\text{C}_9\text{H}_{11}\text{Cl}_3\text{NO}_3\text{PS}$ (Sigma-Aldrich, 99%), and kaolinite clay previously purified. In the chemical analysis were used azomethine-H monosodium salt hydrate - $\text{C}_{17}\text{H}_{12}\text{NNaO}_8\text{S}_2 \cdot x\text{H}_2\text{O}$ (Vetec, 97%) and sodium acetate - $\text{C}_2\text{H}_3\text{NaO}_2 \cdot 3\text{H}_2\text{O}$ (Êxodo Científica, 99%).

Synthesis of the baits

Alginate-kaolinite baits were synthesized using the complex coacervation method adapted from Zhang et al. (2014). The alginate water solution (1% w/v) was prepared under vigorous and constant stirring for 1 hour at 25°C. Then, 2.5 g of kaolinite was added, under stirring for 1 hour at 25°C, and 2.5 g of sodium tetraborate was added to the suspension and stirred for 1 hour at 25°C. The suspension obtained was transferred to a burette and dripped (1 drop/second) into a solution containing 5% CaCl₂ in water (w/v). The material was filtered and dried in an oven at 25 °C for 5 h. The produced bait was named T0.

Another three baits were prepared using new formulations. In this case, the synthesis procedure was the same as described above with some modifications as follows: after preparing the suspension by adding alginate and kaolinite, 2.5 g of sodium tetraborate and 1.0 mL of Boveril[®] were added. Boveril[®] was chosen because it is a biological technique that plays an essential role in ant control (Benjamin et al. 2017). The alginate-kaolinite bait containing sodium tetraborate and Boveril[®] was labeled T1. For the T2 bait synthesis, 5.0 g of sodium tetraborate and 1.0 mL of Boveril[®] containing 1.0 x 10⁸ viable conidia per gram was added to the suspension of alginate and kaolinite. For the synthesis of the T3 bait, the H₂O used to dissolve the alginate was replaced with 600 mg L⁻¹ of the chlorpyrifos insecticide. The same methodology used in the T0 synthesis was followed. Table 1 shows the compositions of the synthesized baits.

Table 1
Compositions of the synthesized bait samples

Bait	H ₂ O	Alginate	Kaolinite	Sodium tetraborate	<i>B. bassiana</i> *	Chlorpyrifos
	(mL)	(g)	(g)	(g)	(mL)	(mL/L)
T0	50	0.5	2.5	2.5	0.0	0.0
T1	50	0.5	2.5	2.5	0.1	0.0
T2	50	0.5	2.5	5.0	0.1	0.0
T3	50	0.5	2.5	2.5	0.0	1000
*Boveril®: 1.0 x 10 ⁸ conidia·g ⁻¹ . WP						

Characterization of materials

The mean diameters of the baits were determined by an average of 10 baits using a digital caliper. X-ray diffraction (XRD) analysis was performed on a Shimadzu XRD-6000 diffractometer, using a graphite crystal monochromator to select Cu-Kα₁ radiation with λ = 1.5406 Å, and a step away from 0.02°, with of 2θ angle between 4° and 70°. Infrared spectra by Fourier transform with attenuated total reflectance (ATR-FTIR) were recorded with a Jasco spectrometer, model FT/IR-4100, in the range of 4000 to 400 cm⁻¹, with 4 cm⁻¹ of resolution and 256 scanings.

Thermogravimetric analyses coupled to differential scanning calorimetry and mass spectrometry (TGA-DSC-MS) were performed using a Netzsch model STA 409 PC – Luxx, coupled to a Netzsch mass spectrometer model QMS 403 C – Aeölos, using a heating range of 10 °C min⁻¹, with N₂ flow (99% N₂) of 50 mL min⁻¹, and a heating range from 25°C temperature to 1000°C. The sample mass was 10 mg.

Prior to scanning electron microscopy (SEM) analyses, the samples were mounted in aluminum specimen holders on a double-sided conductive adhesive and covered with a thin gold film using the Quorum Q150RS equipment. Scanning electron microscopy coupled to energy dispersive X-ray spectroscopy (SEM-EDS) analyses were performed using JEOL JSM-6010/LA instrument with an acceleration voltage of 15 kV.

The concentrations of the B in the baits were determined by UV-V is spectrophotometry using a PerkinElmer Lambda 25 spectrophotometer at a wavelength of 420 nm. Baits were dissolved in HNO₃ and diluted in H₂O. A 1.0 mL aliquot was pipetted into a Falcon tube, and 3.0 mL of H₂O, 1.0 mL of azomethine-H solution (9.0 g L⁻¹), and 1.0 mL of acetate buffer solution were added. The samples were allowed to rest for 30 minutes in a closed environment and protected from light (Reis et al. 2016).

The insecticide chlorpyrifos was quantified using an Agilent Technologies High-Performance Liquid Chromatograph, model 1260 Infinity II, with the ultraviolet detector, model G7117c:225 nm. The chromatographic conditions were as follows: Column C-18, model Eclipse Plus (4.6 mm x 25 cm x 5 µm), Acetonitrile-Water (C₂H₃N:H₂O) in the proportion 80:20 as mobile phase, the temperature of 40 °C, the flow of 1.4 mL min⁻¹ and injection volume of 20 µL.

Boron release experiments

The "*in vitro*" B release experiments were performed in three replicates. First, a bead was suspended in 100 mL of H₂O in an Erlenmeyer flask. Next, this suspension was then placed in a SOLAB SL-223 refrigerated orbital shaker (100 rpm) at 25°C for 48 h. At time intervals of 0.083, 0.25, 0.5, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, and 8.0 hours, 5.0 mL aliquots were extracted and filtered through syringe microfilters (0.45 µm). From this aliquot, 1.0 mL was complexed, as described in section 2.3, and the concentration of B was determined from an analytical curve.

The kinetic study of the "*in vitro*" release of B incorporated into the baits was analyzed using the Higuchi model (Eq. 1), which describes an insoluble matrix as a cumulative percentage of the amount of B released versus the square root of time (Da Costa et al. 2002).

$$Qt = kt^{0.5} \quad (1)$$

Where: Qt = amount of B (mg) released in time (h).

k = Higuchi release constant.

t = time

Boron leaching in soil columns

The B release present in the baits was measured in soil columns using an adapted method from Abat et al. (2015) chemical characteristics of the soil are shown in Table 2. Syringes (columns) with a capacity of 60 mL, a diameter of 3.1 cm, and a height of 17.5 cm were used in this experiment. The bottom of the syringe was sealed with glass wool. It was then filled with 50 g of soil, and the baits were placed at a depth of 5 cm. The system was closed by covering the baits with soil and glass wool. The B rate applied to all treatments was 1.8 kg ha⁻¹. The experiment was conducted in a randomized block design with three replicates. The columns were exposed to H₂O (26 mL) according to the previously determined soil pore volume. The extractions of the leach solutions were performed after 1, 5, 10, and 15 days. A syringe was used to aspirate the solution, and one rubber was attached to the system, connected by a 25 cm long rubber-latex tube. The solutions obtained were collected and packed in plastic containers for analysis by UV-vis spectrophotometry, as described in the 2.3 section.

Table 2
Chemical characterization of a
soil sample.

Attributes	Soil
pH _{H2O}	6.30
P (mg•dm ⁻³)	5.50
Ca ²⁺ (cmol _c •dm ⁻³)	2.20
Mg ²⁺ (cmol _c •dm ⁻³)	0.40
Al ³⁺ (cmol _c •dm ⁻³)	0.29
H + Al (cmol _c •dm ⁻³)	3.70
SB (cmol _c •dm ⁻³)	2.65
t (cmol _c •dm ⁻³)	2.65
T (cmol _c •dm ⁻³)	6.35
V (%)	41.70
m (%)	0.00

P = Mehlich-1 extractant; Ca²⁺, Mg²⁺, and Al³⁺ = KCl 1 mol L⁻¹ extractant; H + Al (potential acidity) = calcium acetate 0.5 mol L⁻¹ extractant – pH 7.0; SB = sum of bases; t = effective cation exchange capacity; T = cation exchange capacity at pH = 7.0; V = base saturation; m = Al saturation index.

Laboratory bioassay

During the assay, the ants were maintained at $25 \pm 1^\circ\text{C}$, $65 \pm 10\%$ RH and photoperiod 12:12 h [L:D]. The leaf-cutting ant workers *A. sexdens* were collected from two colonies located at latitude: $19^\circ 12' 33''$ S and longitude: $46^\circ 07' 55''$ W). The location was a natural pasture with *Brachiaria decumbens* and typical Cerrado shrubs, covering an area of approximately 100 hectares. This area was not treated with insecticides and was far from agricultural areas. Worker ants were collected from two tracks approximately 10 cm wide with a high worker flow. Collections were made between 8:00 and 9:00 a.m. using a fine brush. Approximately 300 ants were collected. Ants were collected and placed in plastic pots (61 x 101 mm) with perforated lids and stored in a B.O.D. (Bio-Oxygen Demand) incubator at a temperature of $25 \pm 1^\circ\text{C}$, relative humidity of $65 \pm 10\%$, and a photoperiod of 12:12 h [L:D]. Five pots containing six ants/pot were used in each series of experiments. Inside the pot, 5.0 cm of clay soil was previously placed, hydrophilic cotton moistened with 1.5 mL of the proposed artificial diet (5% of glucose) dissolved in 100 mL of distilled water, and 15 baits containing the treatments: T0 (control bait with 2.5 g of sodium tetraborate), T1 (bait with 2.5 g of sodium tetraborate and Boveril[®]), T2 (bait with 5.0 g of sodium tetraborate and Boveril[®]), and T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos) were evenly distributed (Bueno et al. 1997). Diets were replaced every 24 h, and the number of dead ants was recorded. Mortality data were corrected for control using the Abbott formula (Abbott 1925).

Field Bioassay

The field bioassay was conducted on pasture (Gramineae) at Donanas Farm (latitude: $19^\circ 13' 15''$ S, longitude: $46^\circ 13' 49''$ W) from June to July 2018, in an area of 50 ha. Anthills were marked using gauged steps, multiplying the greatest width by the most significant length of open land area. Three young anthills were identified, ranging from 1.0 to 4.0 m² for each treatment. Ant baits containing agrochemicals were applied to the total area and consisted of an anthill dosage of 8.0 g m⁻² (Mariconi 1970). The treatments used were: T0 (control bait with 2.5 g of sodium tetraborate), T1 (bait with 2.5 g of sodium tetraborate and Boveril[®]), T2 (bait with 5.0 g of sodium tetraborate and Boveril[®]), T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos), and commercial bait. A total of 15 anthills were evaluated in a completely randomized design with three replications.

The baits were placed approximately 20 cm from the anthill. In each anthill, the number of baits collected and the number of ants m⁻¹ min⁻¹ were evaluated. The number of ants was counted after specific times of 1, 3, and 5 minutes. For all treatments used, the number of baits collected by the ants was evaluated 24 h after the baits were placed on the trails near the anthill. Anthill activity was measured by the frequency of ants per meter per minute (m⁻¹ min⁻¹) on days 0, 14, 35, and 60 at two times, 3:00 pm and 6:00 pm. These times were chosen because they represent the peak foraging activity of the *A. sexdens* species (Giesel et al. 2013).

Statistical analysis

Mortality (%) and ant flow data from the bioassays were subjected to contrast analysis, tested for significance by F-test, and if significant, compared by t-test at $P < 0.05$ using SAS PROC GLM CONTRAST software (SAS 2013). Ant bait collection (%) bioassay data were submitted to analysis of variance and Tukey's test at $p < 0.05$ using SAS PROC ANOVA MEANS software (SAS 2000).

Results

The results of XRD, ATR-FTIR, EDS, and TGA-DSC-MS analyses are presented in the supporting information. The photographic images of the produced baits are shown in Fig. 1. The mean diameters for T0, T1, T2 e T3 are $5,59 \pm 0,35$ mm; $5,13 \pm 0,35$ mm; $5,10 \pm 0,25$ mm, and $5,69 \pm 0,54$ mm, respectively.

Scanning Electron Microscopy (SEM)

Representative images of SEM for the beads are presented in Fig. 2. The morphology of all synthesized beads was quite similar, displaying well-defined and crystalline shapes, as seen in Figs. 2 (a), (c), (e), and (g). Clusters present in Figs. 2 (a), (c), and (e) indicate the presence of monocyclic structures of sodium tetraborate crystals ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$), highlighted by blue circles. In Figs. 2 (a), (e), and (g), the red-highlighted circles denote the presence of the biopolymer in the baits, exhibiting amorphous characteristics. When examining the external morphology of the baits in Figs. 2 (b), (d), (f), and (h), a spherical shape is observed, along with noticeable cracks on its surface.

In vitro release study of boron

The B release from the baits T0, T1, T2, and T3, as a function of time, are shown in Fig. 3. The percentages of B released at 15 minutes, 30 minutes, and 1 hour for T0, T1, T2, and T3 were 55%, 69%, and 98%; 54%, 77%, and 100%; 62%, 87%, and 98%; and 55%, 73%, and 99%, respectively. The experiment showed no significant differences between the release profiles for the different baits, and after 2, 3, 4, 5, 6, 7, and 8 hours, 100% of the B was released for the solution for both baits. The data from the B release study were fitted using the Higuchi's kinetic model, which gave an $R^2 = 0.99$, except for T2, where the R^2 was 0.97. The data from the B release study were fitted using the Higuchi kinetic model, which gave an $R^2 = 0.99$, except for T2 where the R^2 was 0.97.

Boron leaching in soil columns

The results of B leaching in soil columns, as a function of incubation time for $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, T0, T1, T2, and T3, are present in Fig. 4. For $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, 45% of B was leached after 24 h of study, while for T0, T1, T2, and T3, the values were 42%, 22%, 12%, and 44%, respectively. After five days of incubation, $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, T0, T1, T2, and T3 baits leached 57.5%, 48.8%, 80.1%, 58.2% and 49.4% of B, respectively. The T1 and T2 baits leached 100% of B after 10 days, while $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, T0, and T3 leached 61.4%, 56.1%, and 56.1% of B, respectively. At 15 days of incubation, 94.7%, 92.8%, and 91.5% of

total B applied as $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, T0, and T3 baits were recovered in the leachate, respectively. After 15 days, no chlorpyrifos residue was detected in the T3 bait.

Laboratory Bioassay

Figure 5 shows *A. sexdens* mortality percentage as a function of time after contact with the different treatments. The T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos) and commercial bait treatments provided the highest mortality of *A. sexdens* in both treatments, mortality was 100% after 48 h. Treatments T0 (control bait with 2.5 g of sodium tetraborate), T1 (bait with 2.5 g of borate and Boveril®), and T2 (bait with 5.0 g of borate and Boveril®) provided mortality above 75% after 96 h of exposure.

Field Bioassay

In the field bioassay, which was conducted in 15 anthills, the collection of formulated baits and the changes in the behavior of *A. sexdens* ants were analyzed. Figure 6 shows the percentage of bait collection by worker ants in the pasture in the different treatments. Ants collected all formulated baits. The highest percentage of carried baits occurred in treatments T0: control bait with borate, T1: 2.5 g borate and Boveril®, and T2: bait 5.0 g borate and Boveril®, with carried percentages of 100%, 90%, and 85%, respectively. Figure 7 shows the flow of ants in the trails as a function of time after bait release in the different treatments. It can be observed a reduction in the leaf-cutting activity of the worker ants from 14 to 35 days after the release of the baits, except for the treatment T3: bait borate and chlorpyrifos 480 CS, which maintained a low anthill activity up to 60 days after the release.

Discussion

Scanning Electron Microscopy (SEM)

SEM images have revealed the presence of microspheres with diameters ranging between 1200 and 1500 μm , as shown in Fig. 2 (b), (d), (f), and (h). The red circles in Fig. 2 (a and e) show the distribution of the alginate biopolymer present in these baits. Based on Fig. 2 (h), it seems that the sodium tetraborate salt crystals are missing, but the agglomerates imply that there might have been some un-homogenized sodium alginate during the production of the baits. Meanwhile, in Fig. 2 (f), the image of the bait shows more cracks than the others, which could be due to a higher concentration of sodium tetraborate salt used to synthesize this bait. This condition may be linked to a more significant amount of sodium tetraborate salt that was utilized during the synthesis of this bait.

The EDS (Supporting Material) result showed that the baits produced are rich in B. The high content of C and Na may be related due to the presence of the biopolymer sodium alginate formed by mannuronic (M) and guluronic (G) monomers. The high concentration of Cl can be associated with the egg box

process in which the CaCl_2 salt was used to form the crosslinks of the Ca^{2+} ion with carboxyl groups (COO^-) of the alginate. Thus, the Cl is retained in the system (De Castro et al. 2020).

The egg box system in the alginate polymer, combined with kaolinite, acts as a protective barrier in the beads. This barrier effectively reduces the diffusion rate of boron into the beads, indicating that these beads could be a feasible controlled transport option for active agents. This is particularly significant given the importance of the controlled release of active compounds in industries such as pharmaceuticals and agriculture. Additionally, the high concentration of Cl observed in the beads can be attributed to the use of CaCl_2 salt in the crosslinking process, which ultimately contributes to the formation of beads.

In vitro release study of boron

The B release study for baits T0, T1, T2, and T3 resulted in a similar B release profile (Fig. 3). In another study using the same organic matrix (alginate) that was used in the present research, Castro et al. conducted a study with different sources of B fertilizers, including boric acid and Ulexite, which are commercial sources with high and low-solubility, respectively (Mi et al. 1997). The authors compared the commercial sources with the alginate microspheres containing boric acid (BA-ALG) and layered double hydroxides intercalated with borate (LDH-B-ALG). The B release in the solution showed that 100% of the B was released after 1, 2, 6, and 24 hours of application of boric acid, BA-ALG, Ulexite, and LDH-B-ALG, respectively. The B present in boric acid showed a faster release rate when compared to the other sources due to the solubilization of this acid. In the case of the present study, the sodium tetraborate salt ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) present in the baits could be released faster due to the high solubility in H_2O ($2.7 \text{ g } 100 \text{ mL}^{-1}$). All baits remained in their initial physical form after 8 h, indicating that the H_2O is not a destructive factor for the baits under the conditions of this study. It is noteworthy that commercial baits lose their shape in wet environments and that the shape of the baits is essential for the ants to carry them to the anthill. These results show the importance of the interaction between the pristine parts of the organic-inorganic hybrid to synthesize new materials that can be an alternative bait with a more gradual release of B, which in this case of interest, can increase the control efficiency of ants.

The results from Higuchi's kinetic model in the 3.2 section presented the $R^2 = 0.97$ for T2 and $R^2 = 0.99$ for T0, T1, and T3. All treatments showed a sustained release profile up to the first hour of the study ($t_{0.5} = 1$). (Higuchi 1961, 1963, Mi et al. 1997, De Paula et al. 2010) After this time, the equilibrium was reached with a constant release rate up to 8 h ($t_{0.5} = 6.92$). Higuchi's model is based on Fick's law of diffusion, so the release of the B from the baits occurred by diffusion, in which the H_2O enters the baits through cracks and by pores via a spontaneous process, initiating competition between the matrix and the H_2O by the agrochemical. In this process, as the bait presents a higher amount of solute, the micronutrient moves to the region with a lower concentration until the chemical equilibrium is reached.

Boron leaching in soil columns

The baits containing *B. Bassiana* showed a higher B leaching profile at 10 days of incubation. In general, the results showed a higher B leaching profile from $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ compared to baits T0 and T3. Previous research reported a lower B leaching in soil columns from LDH-B-ALG and BA-ALG compared to acid boric (De Castro et al. 2020) These results confirm that alginate polymers combined with materials such as clays and agrochemicals can provide physical protection and reduce the rate of H_2O diffusion into the baits, which can lead to lower B release. The slow leaching profile of the baits is an important feature to increase the residence time of B in the environment, thus avoiding losses to soil and groundwater contamination. This slow-release profile can allow the use of lower rates and reduce the frequency of active ingredient (insecticide) applications for leaf-cutting ants control.

For the bait that contained chlorpyrifos, no active chemical was detected at 15 days of incubation, possibly because this insecticide has low water solubility and a high sorption rate in soils (Mi et al. 1997). During the foraging cycle performed by ants, the baits taken into the anthill have ant-killing effects. It is also worth noting that these baits may have a secondary function as a B fertilizer due to the presence of B, as this chemical element is a micronutrient for growing plants.

Laboratory Bioassay

Based on the experiment, it was observed that on days 2, 3, and 4, treatments T3 and T4 showed the lowest means. However, it was noted that the T3 treatment had 100% mortality after 24 h of evaluation. After 120 h of the experiment, all baits displayed efficient ant-killing activity. Therefore, it can be inferred that these baits are effective in the chemical control of these insects. It is interesting to note that the T3 bait had better results than the bait T4, which is one of the most commercial baits used in Brazil for ant control. This superiority may be due to the insecticidal action of chlorpyrifos.

Insects are affected by chlorpyrifos by interfering with the transmission of nerve impulses, which leads to paralysis and, eventually, death. When the insecticide is ingested or comes into contact with the insect, a process occurs where the chlorpyrifos binds to the esteratic center of acetylcholinesterase (AChE). This causes a cholinergic syndrome that prevents the hydrolysis of the neurotransmitter (ACh). Commercial baits containing sulfluramid insecticide are also effective in inhibiting the metabolic process of oxidative phosphorylation and acting on the respiratory chain of insects. In the mitochondria, for example, the flow of electrons is blocked, interrupting the production of adenosine triphosphate (ATP). This affects the energy storage process for vital activities, leading to asphyxia and the death of the insects (Higuchi 1961, 1963).

Field Bioassay

Treatment T1 showed the highest averages, unlike what happens with bait T2, which has the same composition as bait T1, except for the amount of sodium tetraborate. Therefore, the difference in the results of “on the indecision” between the T1 and T2 baits may be related to the more significant amount of tetraborate present in the T2 bait. This amount of salt may have contributed to minimizing/inhibiting the odor of *B. bassiana* spores in these baits. The baits T0, T2, T3, and T4 had potential attractiveness

since, after the first contact, the insects showed good acceptance. The ants are not repelled by the added ingredients in the baits, including the toxic ingredient. The greater acceptance of the baits by *A. sexdens* is probably because the baits are odorless and do not have an acute toxic effect on them since the baits do not cause the immediate death of the ants upon contact, allowing them to carry the baits into the interior of the anthill. Interestingly, *A. sexdens* are particularly attracted to these baits, suggesting that they may be particularly effective against this ant species.

The ant flow variable aimed to evaluate the number of insects that entered the anthill in a 5-minute time interval during the evaluation days. As the anthills were not opened, one way to assess the mortality of these insects was to count the decrease in the flow of ants during the evaluation days after the application of the baits. If there is a decrease in foraging activity (flow of ants), an external factor likely causes the mortality of these insects. Thus, the objective was to evaluate the flow of ants to determine the mortality of these insects. When comparing the synthesized baits with the commercial ones already consolidated in the market (Britto et al. 2016, Mota Filho et al. 2021), the bioassay in the field using the ant flow variable did not indicate a significant difference. Therefore, the efficacy of all baits produced can be confirmed.

Results shown in Fig. 7 support the information obtained for the variable flow of ants, where a decrease in foraging activity was observed, which was probably caused by the death of these insects. Since the baits were collected and transported into the anthill, this shows that the active ingredients in the baits did not have an acute lethal effect because the baits did not kill the ants during transport, which can provide a more significant lethal effect, as it affects a greater number of individuals that are inside the anthill. Information reported in the literature shows that when ants collect insecticide baits, they do not perform trophallaxis, and transport the baits inside the anthill to the chamber where the fungi are located, where contamination of the colony occurs (Mota Filho et al. 2021, De Andrade et al. 2002, Camargo et al. 2017). This fact is strengthened by the knowledge that it is not necessary to release many baits to change the behavior of the colonies (Forti et al. 2019). These results strongly suggest that the baits produced in this work can be considered a new option for ant control without presenting the shape deformation or loss of effectiveness very common to the commercial ones when these are submitted to humid climates.

Conclusion

New baits for leaf-cutting ants control were developed by incorporating sodium tetraborate, *B. bassiana* spores, and chlorpyrifos into alginate-kaolinite beads. The success of this incorporation was confirmed through various techniques, including DRX, FTIR-ATR, SEM-EDS, and TGA-DSC-MS, which provided insight into the crystallinity, composition, structure, and thermal stability of the baits. In the "*in vitro*" release tests, there was a 100% release of B in the baits during the first hour of the study. This release occurred through a diffusion process. The leaching of B from the baits was faster for T1 and T2 as compared to Na₂B₄O₇·10H₂O, T0, and T3. The synthesized baits were found to be highly efficient in managing leaf-cutting ants of the genus *A. sexdens* both in laboratory and field tests. Moreover, they

were also found to be very attractive to these ants. Of all the baits created, the T3 bait had the highest potential to be used in the management of *A. sexdens* ants. Furthermore, it had the lowest rejection averages and was the most effective in killing insects. These innovative baits offer a promising solution for controlling leaf-cutting ants.

Declarations

Acknowledgments

The present work was carried out with the support of the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - Financing Code 001, Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG), process numbers APQ-02360-18, APQ-03410-22, and RED-00056-23. The authors are also grateful to the Rede Mineira de Química (RQ-MG) and Programa de Pós-Graduação Multicêntrico em Química de Minas Gerais (PPGMQ-MG).

Conflict of interest declaration

All the authors confirmed that there is no conflict of interest.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting information

Supporting information may be found in the online version of this article.

Author Contribution

Valber G. de O. Duarte performed all experiments, including bait synthesis, boron release experiments, boron leaching in soil columns, laboratory bioassay, and field bioassay. Valber G. de O. Duarte also wrote the manuscript. Victor dos S.A. Leite participated in the synthesis and characterization of the beads. Danúbia A. da C. Nobre contributed to the statistical analyses. Vera R.L. Constantino performed and analyzed the TGA-DSC-MS and FTIR analyses. Renê C. da Silva performed and analyzed the SEM analyses. Gustavo F. de Castro contributed to and analyzed the boron release experiments and boron leaching in soil columns. Flávio L. Fernandes designed, performed, and analyzed the laboratory and field bioassay experiments. Flávio L. Fernandes also contributed to the drafting of the manuscript. Jairo Tronto oversaw all manuscript-related processes.

References

1. Abat M, Degryse F, Baird R, McLaughlin MJ (2015) Slow-release boron fertilisers: co-granulation of boron sources with mono-ammonium phosphate (MAP). *Soil Res* 53:505–551. <https://doi.org/10.1071/SR14128>
2. Abbott WS (1925) A method of computing the effectiveness of an insecticide. *J Econ Entomol* 18:265–267.
3. Abellán G, Martí-Gastaldo C, Ribera A, Coronado E (2015) Hybrid materials based on magnetic layered double hydroxides: a molecular perspective. *Acc Chem Res* 48:1601–1611. <https://doi.org/10.1021/acs.accounts.5b00033>
4. Benício LPF, Silva RA, Lopes JA, Eulálio D, Santos RMM, Aquino LA et al (2015) Layered double hydroxides: nanomaterials for applications in agriculture. *Rev Bras Ciec Solo* 39:1–13. <https://doi.org/10.1590/01000683rbcscs2015081>
5. Benjamin L, Sevugapperumal N, Kennedy J, Manoharan T (2017) Morphological and molecular characterization of entomopathogenic fungi, isolated from different *Beauveria bassiana* insects in India. *Green Farming* 4:940-944.
6. Boulogne I, Ozier-lafontaine H, Loranger-Merciris G (2014) Leaf-cutting ants, biology and control, in sustainable agriculture reviews. Springer
7. Britto JS De, Forti LC, Oliveira MA De, Zanetti R, Wilcken CF, Zanuncio JC, et al (2016) Use of alternatives to PFOS, its salts and PFOSF for the control of leaf-cutting ants *Atta* and *Acromyrmex*. 3:11–92.
8. Bueno OC, Morini MSC, Pagnocca FC, Hebling MJA, Silva OA (1997) Sobrevivência de operárias de *Atta sexdens rubropilosa* Forel (Hymenoptera: Formicidae) isoladas do formigueiro e alimentadas com dietas artificiais. *An da Soc Entomológica do Bras* 26:107–113. <https://doi.org/10.1590/S0301-80591997000100015>
9. Buratto DA, Sousa NJ, Souza MD, Rolin FA (2017) Resistência de iscas granuladas, distribuídas a granel e em microporta-iscas, à ação da umidade em plantio de *Pinus taeda* no planalto sul-catarinense. *Cienc Florest* 27:1083–1093. <https://doi.org/10.5902/1980509828683>
10. Camargo RS, Puccini C, Forti LC (2017) Allogrooming, self-grooming, and touching behavior: contamination routes of leaf-cutting ant workers using a fat-soluble tracer dye. *Insects* 8:59. <https://doi.org/10.3390/insects8020059>
11. Cantú-Ruiz AL, Galván-Quintero AO, Mar-Solis LM (2017) Aplicaciones biotecnológicas en el control biológico. *Artrópod y Salud* 7:54-70.
12. Cao L, Lu W, Mata A, Nishinari K, Fang Y (2020) Egg-box model-based gelation of alginate and pectin: a review. *Carbohydr Polym* 242:116389. <https://doi.org/10.1016/j.carbpol.2020.116389>
13. Chan LW, Jin Y, Heng PWS (2002) Crosslinking mechanisms of calcium and zinc in production of alginate microspheres. *Int J Pharm* 242:255-258. [https://doi.org/10.1016/s0378-5173\(02\)00169-2](https://doi.org/10.1016/s0378-5173(02)00169-2)
14. Cruz AP, Zanuncio JC, Zanetti R, Gomes OS (1996) Eficiência de iscas granuladas a base de sulfluramida e de clorpirifós no controle de *Atta sexdens sexdens* (Hymenoptera: Formicidae), no trópico úmido. *Acta Amaz* 26:145–150. <https://doi.org/10.1590/1809-43921996263150>

15. Cunha VRR, Ferreira CAM, Constantino RLV (2010) Hidróxidos duplos lamelares: nanopartículas inorgânicas para armazenamento e liberação de espécies de interesse biológico e terapêutico. Quim Nova 33:159–171. <https://doi.org/10.1590/S0100-40422010000100029>
16. Da Costa PJC (2002) Avaliação in vitro da lioequivalência de formulações farmacêuticas. Rev Bras Ciencias Farm J Pharm Sci 38:141–153. <https://doi.org/10.1590/S1516-93322002000200003>
17. Darder M, Lo M, Aranda P, Leroux F, Ruiz-hitzky E, September RV et al. (2005) Bio-nanocomposites based on layered double hydroxides. 17:1969–1977. <https://doi.org/10.1021/cm0483240>
18. De Andrade APP, Forti LC, Moreira AA, Boaretto MAC, Ramos VM, De Matos CAO (2002) Behavior of *Atta sexdens rubropilosa* (Hymenoptera: Formicidae) workers during the preparation of the leaf substrate for symbiont fungus culture. Sociobiology 40:293–306.
19. De Castro GF, Zotarelli L, Mattiello EM, Tronto J (2020) Alginate beads containing layered double hydroxide intercalated with borate: a potential slow-release boron fertilizer for application in sandy soils. New J Chem 44:16965–16976. <https://doi.org/10.1039/d0nj03571h>
20. De Paula HCB, De Oliveira EF, Abreu FOMS, De Paula RCM, De Moraes SM, Forte MMC (2010) Esferas (beads) de alginato como agente encapsulante de óleo de croton zehntneri Pax et Hoffm. Polimeros 20:112–120. <https://doi.org/10.1590/S0104-14282010005000019>
21. Della Lucia TMC (2011) Formigas cortadeiras da bioecologia ao manejo. Editora UFV, Viçosa.
22. Etchepare MDA, De Menezes MFC, Rodrigues LZ, Codevilla CF, De Menezes CR (2015) Microencapsulation of bioactive compounds by extrusion method. Ciência e Nat 37:97–105. <https://doi.org/10.5902/2179460X19725>
23. Fernandes FM, Baradari H, Sanchez C (2014) Integrative strategies to hybrid lamellar compounds: an integration challenge. Appl Clay Sci 100:2–21. <https://doi.org/10.1016/j.clay.2014.05.013>
24. Folgarait PJ, Goffré D, Osorio AG (2020) *Beauveria bassiana* for the control of leafcutter ants: strain and host differences. Biocontrol Sci Technol 30:996–1005. <https://doi.org/10.1080/09583157.2020.1772199>
25. Forti LC, Camargo RS, Andrade APP, Catalani GC, Sousa KKA, Silva AAC, et al (2019) Contamination route of leaf-cutting worker ants analyzed through a fat-soluble tracer dye in toxic bait. Neotrop Entomol 48:349–355. <https://doi.org/10.1007/s13744-018-0652-1>
26. Forti LC, Pretto DR, Nagamoto NS, Padovani CR, Camargo RS, Andrade APP (2007) Dispersal of the delayed action insecticide sulfluramid in colonies of the leaf-cutting ant *Atta sexdens rubropilosa* (Hymenoptera: Formicidae). Sociobiology 50:1149–1163.
27. Furtado EL, Jesus Júnior WC, Moraes WB (2013) Forest diseases in Brazil: status and management. In: Estay SA (ed) Forest pest and disease management in Latin America, 1st edn. Springer, New York, pp 211–230.
28. Giesel A, Boff MIC, Boff P (2013) Seasonal activity and foraging preferences of the leaf-cutting ant *Atta sexdens piriventris* (Santschi) (Hymenoptera: Formicidae). Neotrop Entomol 42:552–557. <https://doi.org/10.1007/s13744-013-0160-2>

29. Hanna R, Fotso Kuate A, Nanga SN, Tindo M and Nagel P (2015) Boric acid for suppression of the ant *Anoplolepis tenella* and effects on an associated scale insect pest *Stictococcus vayssierei* in cassava fields in the Congo Basin. *Crop Prot* 74:131–137.
<https://doi.org/10.1016/j.cropro.2015.04.016>
30. Haug A, Smidrod O (1970) Selectivity of some anionic polymers for divalent metal ions. *Acta Chem Scand* 24:843–854. <https://doi.org/10.3891/ACTA.CHEM.SCAND.24-0843>
31. Higuchi T (1963) Mechanism of sustained-action medication. Theoretical analysis of rate of release of solid drugs dispersed in solid matrices. *J Pharm Sci* 52:1145–1149.
<https://doi.org/10.1002/jps.2600521210>
32. Higuchi T (1961) Rate of release of medicaments from ointment bases containing drugs in suspension. *J Pharm Sci* 50:874–875. <https://doi.org/10.1002/jps.2600501018>
33. Hooper-Bui LM, Rust MK (2000) Oral toxicity of abamectin, boric acid, fipronil, and hydramethylnon to laboratory colonies of argentine ants (Hymenoptera: Formicidae). *J Econ Entomol* 93:858–864.
<https://doi.org/10.1603/0022-0493-93.3.858>
34. Jain S, Datta M (2016) Montmorillonite-alginate microspheres as a delivery vehicle for oral extended release of venlafaxine hydrochloride. *J Drug Deliv Sci Technol* 33:149–156.
<https://doi.org/10.1016/j.jddst.2016.04.002>
35. Klotz JH, Greenberg LES, Amrhein C, Rust MK, Greenberg LES (2000) Toxicity and repellency of borate-sucrose water baits to argentine ants (Hymenoptera: Formicidae). *J Econ Entomol* 93:1256–1258. <https://doi.org/10.1603/0022-0493-93.4.1256>
36. Li ZZ, Huang B, Chen MJ, Wang B, Fan MZ (2011) Studies on the genus *Beauveria* in molecular era. *Mycosystema* 30:823–835.
37. Lima ACS, Souza CZF, Oliveira AHC, Alves JMA, Correia RG (2011) Diagnóstico fitossanitário e de práticas associadas ao uso de agrotóxicos nas hortas em ambiente protegido em Boa Vista – Roraima. *Rev Agroambiente* 5:24-133. <https://doi.org/10.18227/1982-8470ragro.v5i2.482>
38. Loureiro ES, Monteiro AC (2005) Patogenicidade de isolados de três fungos entomopatogênicos a soldados de *Atta sexdens sexdens* (Linnaeus, 1758) (Hymenoptera: Formicidae). *Rev Árvore* 29:553–561. <https://doi.org/10.1590/S0100-67622005000400007>
39. Loureiro ES, Busarello GD, Pessoa LGA, Uchoa-Fernandes MA, Amaral TS, Pessoa MB (2022) Pathogenicity of *Beauveria bassiana* to *Atta sexdens rubropilosa* (Hymenoptera: Formicidae) in laboratory conditions. *Res Soc Dev* 11: e41011225831. <https://doi.org/10.33448/rsd-v11i2.25831>
40. Mariconi FAM (1970) As Saúvas. Ceres, São Paulo.
41. Mi F-L, Tsung-Bi W, Shyu S-S (1997) Sustained-release of oxytetracycline from chitosan microspheres prepared by interfacial acylation and spray hardening methods. *J Microencapsul* 14:577–591. <https://doi.org/10.3109/02652049709006811>
42. MAPA (2003) Agrofit- Sistema de Agrotóxico Fitossanitários.
https://agrofit.agricultura.gov.br/agrofit_cons/principal_agrofit_cons. Accessed 26 July 2023.

43. Moreira AA, Forti LC, Andrade APP, Boaretto MAC, Lopes JFS (2004) Nest architecture of *Atta laevigata* (F. Smith, 1858) (Hymenoptera: Formicidae). *Stud Neotrop Fauna Environ* 39:109–116. <https://doi.org/10.1080/01650520412331333756>
44. Mota Filho TMM, Camargo RS, Zanuncio JC, Stefanelli LEP, de Matos CAO, Forti LC (2021) Contamination routes and mortality of the leaf-cutting ant *Atta sexdens* (Hymenoptera: Formicidae) by the insecticides fipronil and sulfluramid through social interactions. *Pest Manag Sci* 77:4411–4417. <https://doi.org/10.1002/ps.6475>
45. Nakagawa LE, Potenza MR, Ichirraro TH, Papini S (2016) Persistence of chlorpyrifos in greenhouse air. *Arq Inst Biol* 83:1-4. <https://doi.org/10.1590/1808-1657001052014>
46. Peregrine DJ, Cherrett JM (1976) Toxicant spread in laboratory colonies of the leaf-cutting ant. *Ann Appl Biol* 84:128–133. <https://doi.org/10.1111/j.1744-7348.1976.tb01742.x>
47. Pérez SP, Corley JC, Farji-Brener AG (2011) Potential impact of the leaf-cutting ant *Acromyrmex lobicornis* on conifer plantations in northern Patagonia, Argentina. *Agric For Entomol* 13:191–196. <https://doi.org/10.1111/j.1461-9563.2010.00515.x>
48. Reis C, Reis EL, Silva CA (2006) Modificação de procedimentos para extração de boro em solos. *Recie* 15:91–94.
49. Reis M de A, Cunha JPAR, Zanetti R, Fernandes BV, Reis JMR (2015) Aplicação sistemática mecanizada de isca formicida granulada em eucaliptais em fase de manutenção. *Cerne* 21:423–428. <https://doi.org/10.1590/01047760201521031331>
50. Sanchez C, Shea KJ, Kitagawa S (2011) Recent progress in hybrid materials science. *Chem Soc Rev* 40:471–472. <https://doi.org/10.1039/C1CS90001C>
51. Sanchez C, Shea KJ, Kitagawa S, Schubert U (2011) Hybrid materials themed issue Cluster-based inorganic-organic hybrid materials. *Chem Soc Rev* 40:453–1152.
52. SAS Institute (2013). Statistical analysis system: getting started with the SAS learning. SAS Institute: Cary, NC, USA.
53. Singh BK, Walker A (2006) Microbial degradation of organophosphorus compounds. *FEMS Microbiol Rev* 30:428–471. <https://doi.org/10.1111/j.1574-6976.2006.00018.x>
54. Sola F, Falibene A, Josens R (2013) Asymmetrical behavioral response towards two boron toxicants depends on the ant species (Hymenoptera: Formicidae). *J Econ Entomol* 106:929–938. <https://doi.org/10.1603/ec12246>
55. Suiter DR, Gochnour BM, Holloway JB, Vail KM (2021) Alternative methods of ant (Hymenoptera: Formicidae) control with emphasis on the argentine ant, *Linepithema humile*. *Insects* 12:487. <https://doi.org/10.3390/insects12060487>
56. Swanson AC, Schwendenmann L, Allen MF, Aronson EL, Artavia-León A, Dierick D et al (2019) Welcome to the Atta world: a framework for understanding the effects of leaf-cutter ants on ecosystem functions. *Func Ecol* 33:1386-1399. <https://doi.org/10.1111/1365-2435.13319>
57. Takahashi T, Yamaguchi M (1991) Host-Guest interaction between swelling clay minerals and poorly water-soluble drugs. 1: complex formation between a swelling clay mineral and griseofulvin. *J Incl*

58. Teixeira F, Souza T, Castellani M, Camargo R, Forti L, Paula V et al (2023) Bioactivity of plant extracts against *Atta sexdens* (Hymenoptera: Formicidae) workers by ingestion. Int J Trop Insect Sci 43. <https://doi.org/10.1007/s42690-023-01030-6>
59. Urbas P, Araujo Júnior MV, Leal IR, Wirth R (2007) Cutting more from cut forests: drastic edge effects on colony density and herbivory pressure of leaf-cutting ants in Brazil. Biotropica 39:489–495. <https://doi.org/10.1111/j.1744-7429.2007.00285.x>
60. Vieira EFS, Cestari AR, Chagas RA, Cortes GKDR (2014) Obtenção e caracterização de matriz apropriada para sistemas de liberação prolongada-estudos de liberação dos herbicidas atrazina e diuron. Quim Nova 37:398–403. <https://doi.org/10.5935/0100-4042.20140074>
61. Wan LSC, Heng PWS, Chan LW (1994) Surfactant effects on alginate microspheres. Inrernariunul J Pharm 103:267–275. [https://doi.org/10.1016/0378-5173\(94\)90177-5](https://doi.org/10.1016/0378-5173(94)90177-5)
62. Wołejko E, Łozowicka B, Jabłońska-Trypuć A, Pietruszyńska M, Wydro U (2022) Chlorpyrifos occurrence and toxicological risk assessment: a review. Int J Environ Res Public Health 19:19. <https://doi.org/10.3390/ijerph191912209>
63. Zanetti R, Zanuncio JC, Mayhé-Nunes AJ, Medeiros AGB, Souza-Silva A (2003) Combate sistemático de formigas-cortadeiras com iscas granuladas, em eucaliptais com cultivo mínimo. Rev Árvore 27:387–392. <https://doi.org/10.1590/S0100-67622003000300016>
64. Zanetti R, Zanuncio JC, Vilela EF, Leite HG, Jaffé K, Oliveira AC (2003) Level of economic damage for leaf-cutting ants (Hymenoptera: Formicidae) in eucalyptus plantations in Brazil. Sociobiology 42:433–442.
65. Zanúncio JC, Couto L, Santos GP and Zanúncio TV (1992) Eficiência da isca granulada à base de sulfamid no controle da formiga cortadeira *Atta laevigata* (F. Smith, 1858) (Hymenoptera: Formicidae). Rev Arvore 16: 357–361.
66. Zhang Y, Liang X, Yang X, Liu H, Yao J (2014) An eco-friendly slow-release urea fertilizer based on waste mulberry branches for potential agriculture and horticulture applications. ACS Sustain Chem Eng 2:1871–1878. <https://doi.org/10.1021/sc500204z>

Figures

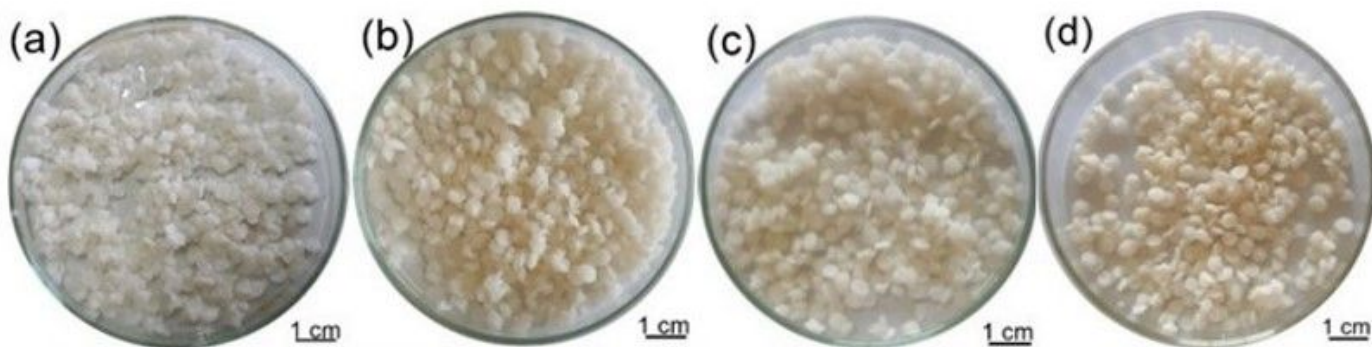


Figure 1

Photographic images of the produced baits: (a) T0 (control bait with 2.5 g of sodium tetraborate), (b) T1 (bait with 2.5 g of sodium tetraborate and Boveril[®]), (c) T2 (bait with 5.0 g of sodium tetraborate and Boveril[®]), and (d) T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos).

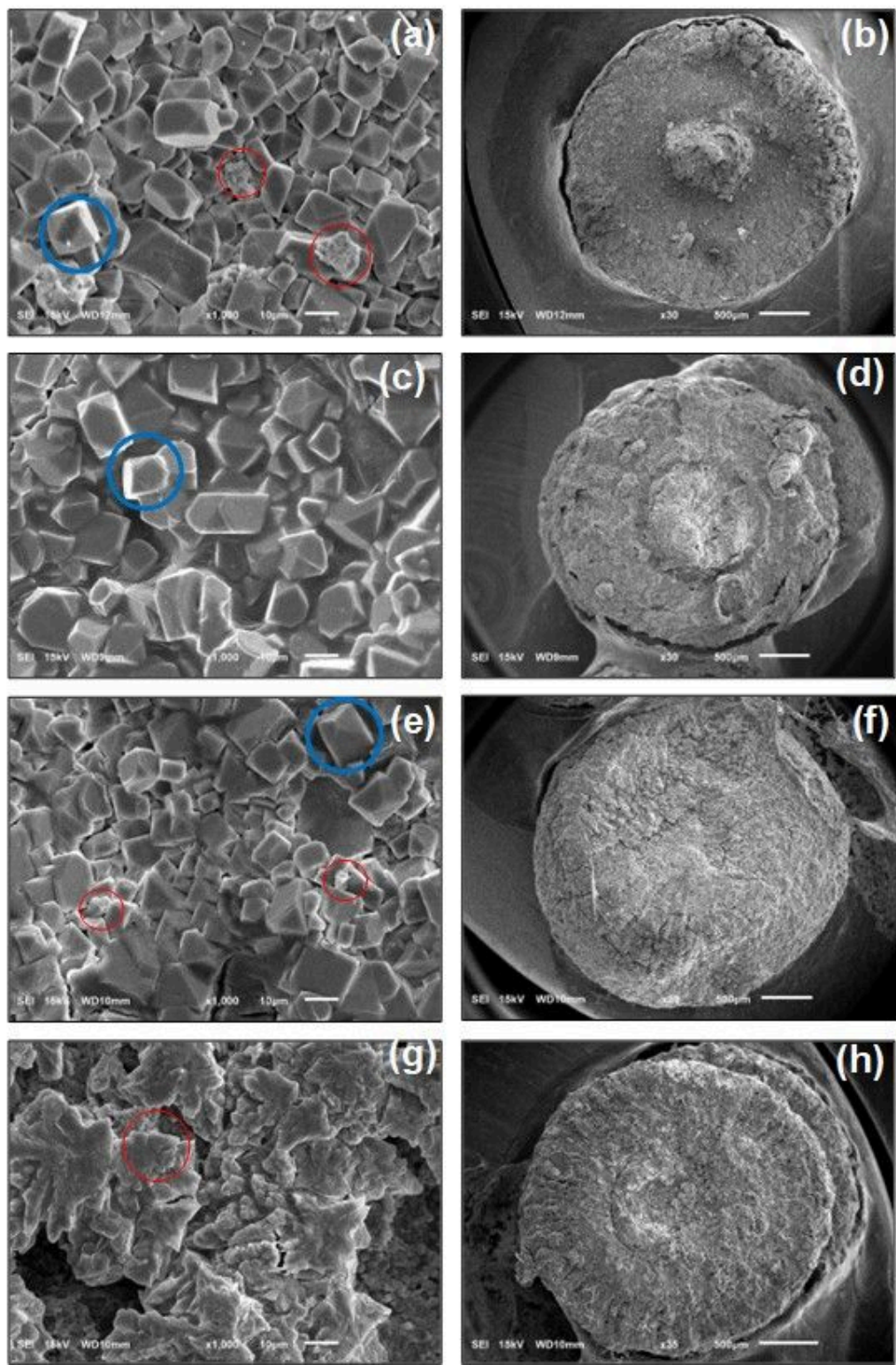


Figure 2

Representative images of scanning electron microscopy coupled to X-ray dispersive spectroscopy (SEM-EDS) of the samples: (a) and (b) T0 (control bait with 2.5 g of sodium tetraborate); (c) and (d) T1 (bait with 2.5 g of sodium tetraborate and Boveril[®]); (e) and (f) T2 (bait with 5.0 g of sodium tetraborate and Boveril[®]); (g), and (h) T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos). Left figures: high magnifications. Right figures: low magnifications.

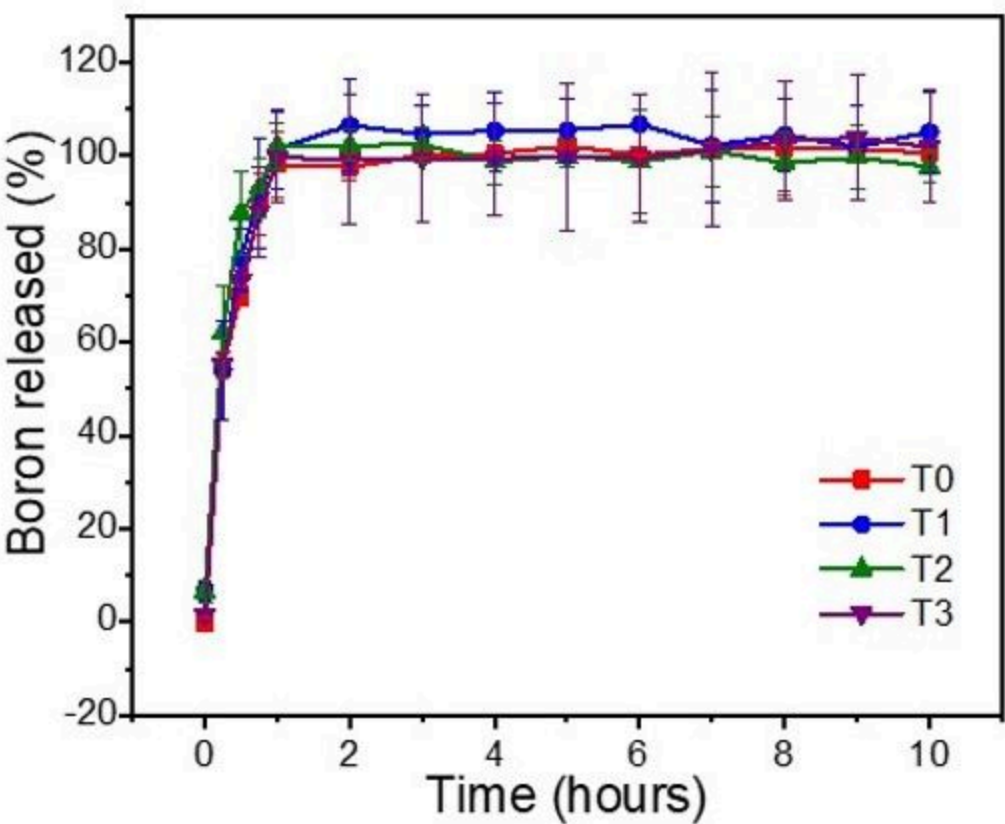


Figure 3

Release of boron in solution as a function of time in "*in vitro*" release studies for T0 (control bait with 2.5 g of sodium tetraborate), T1 (bait with 2.5 g of sodium tetraborate and Boveril[®]), T2 (bait with 5.0 g of sodium tetraborate and Boveril[®]), and T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos).

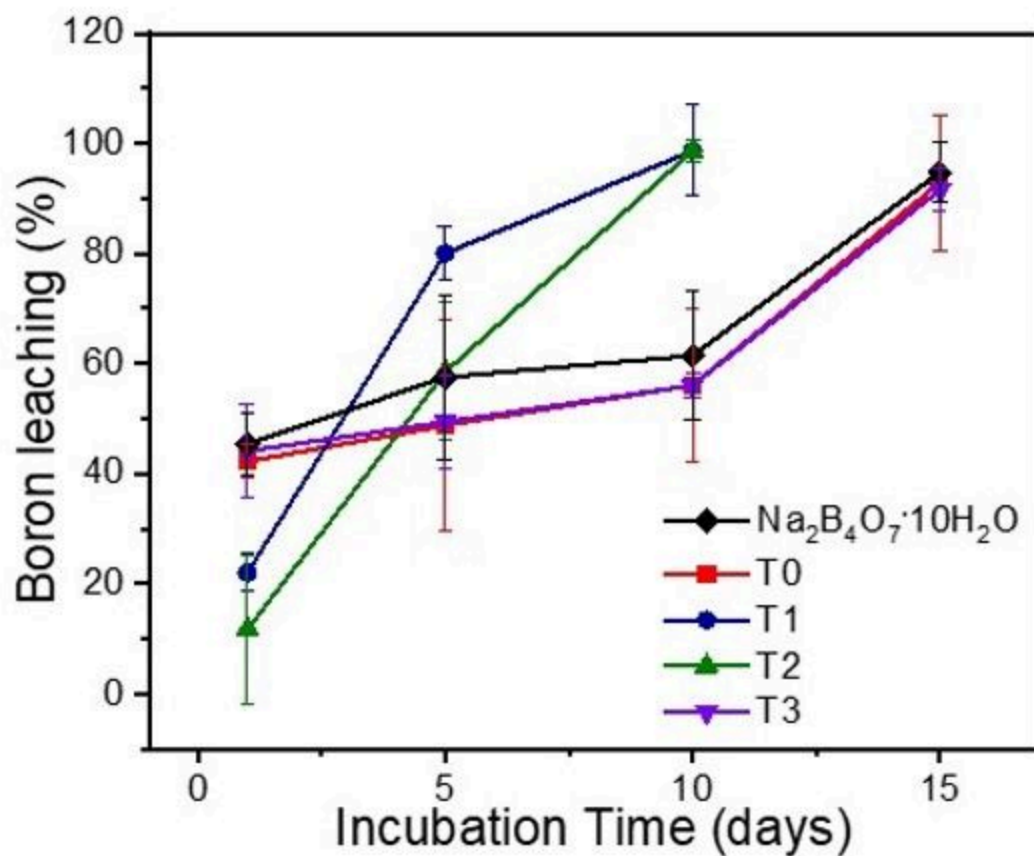


Figure 4

Boron leaching study in soil columns as a function of incubation time for $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, T0 (control bait with 2.5 g of sodium tetraborate), T1 (bait with 2.5 g of sodium tetraborate and Boveril[®]), T2 (bait with 5 g of sodium tetraborate and Boveril[®]), and T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos).

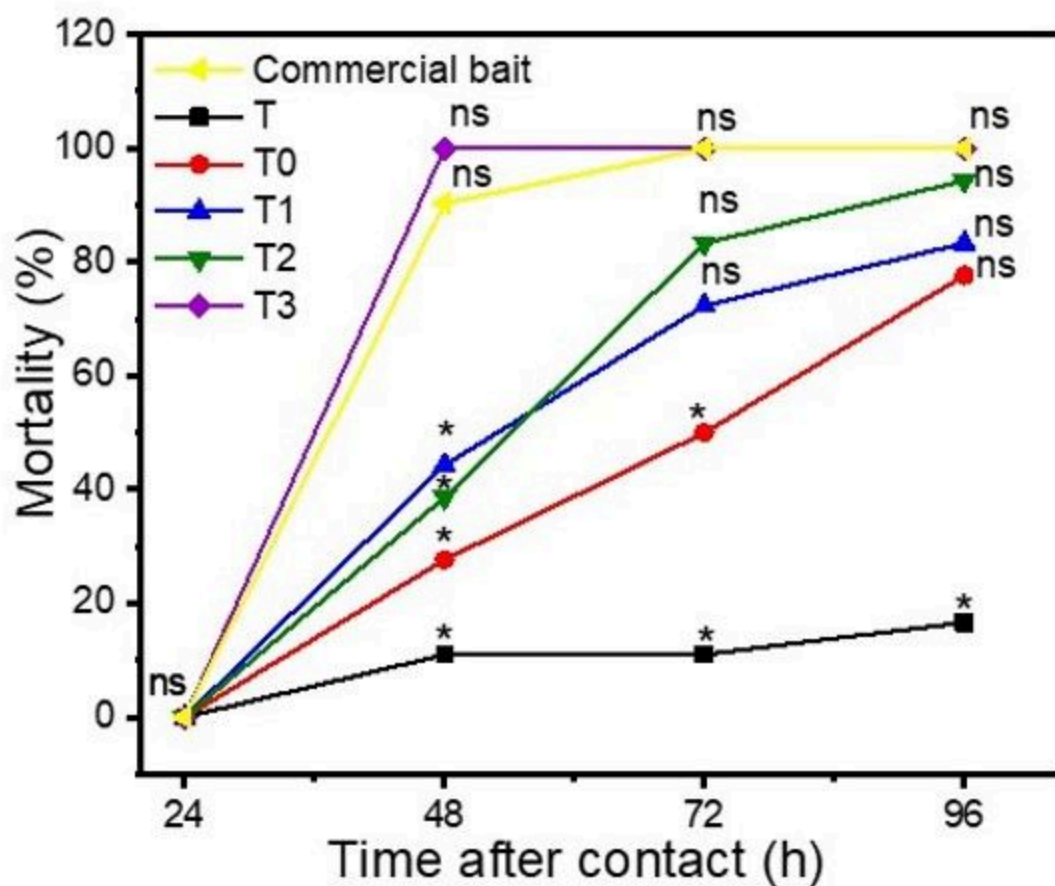


Figure 5

Mortality of *A. sexdens* at 24, 48, 72, and 96 hours after contact with baits. Treatments: T(Control), T0 (control bait with 2.5 g of sodium tetraborate), T1 (bait with 2.5 g of borate and Boveril®), T2 (bait with 5.0 g of borate and Boveril®), T3 (bait with 2.5 g of sodium tetraborate and chlorpyrifos), and commercial bait. In the treatments, the mortality of the ants was corrected by the control using the Abbott formula (1925). Symbol (*) shows a significant difference, ns= not significant (T test, $P < 0.05$).

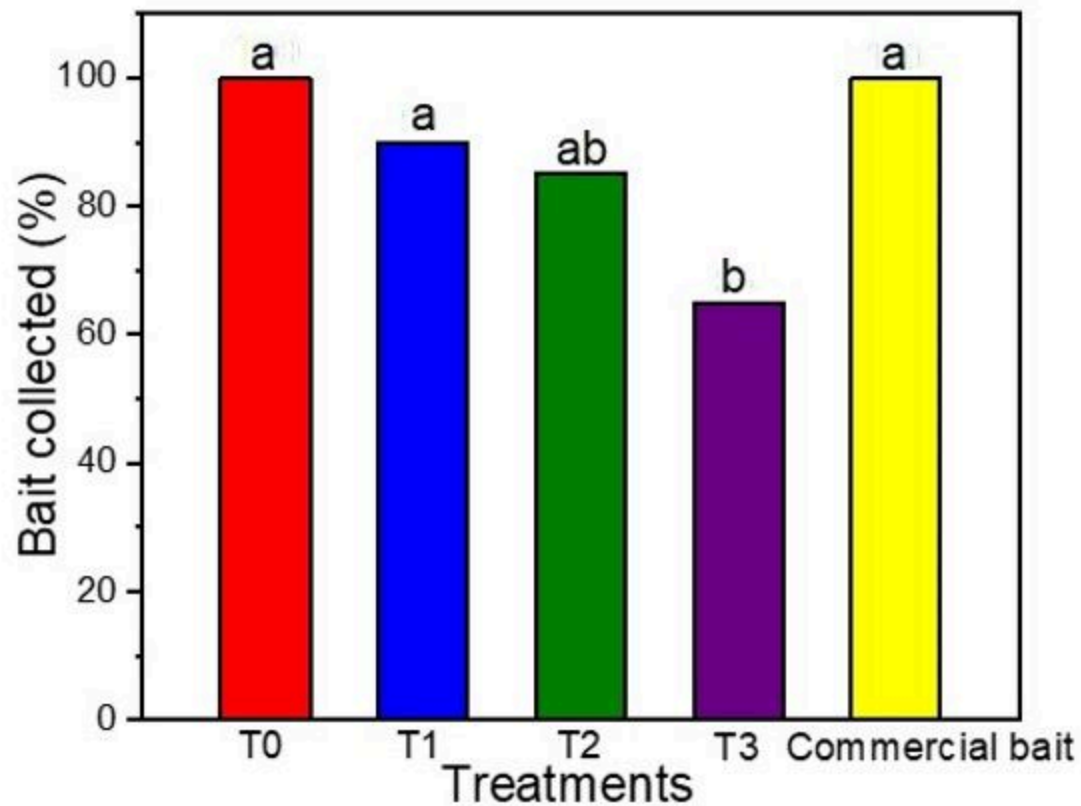


Figure 6

Percentage of bait collected by worker ants in the pasture in different treatments. Treatments - T0: control bait with 2.5 g of sodium tetraborate, T1: bait with 2.5 g sodium tetraborate and Boveril[®], T2: bait with 5.0 g borate and Boveril[®], T3: bait with 2.5 g of sodium tetraborate and chlorpyrifos and commercial bait. Means followed by the same letter in the bar are not different by Tukey's test ($P < 0.05$).

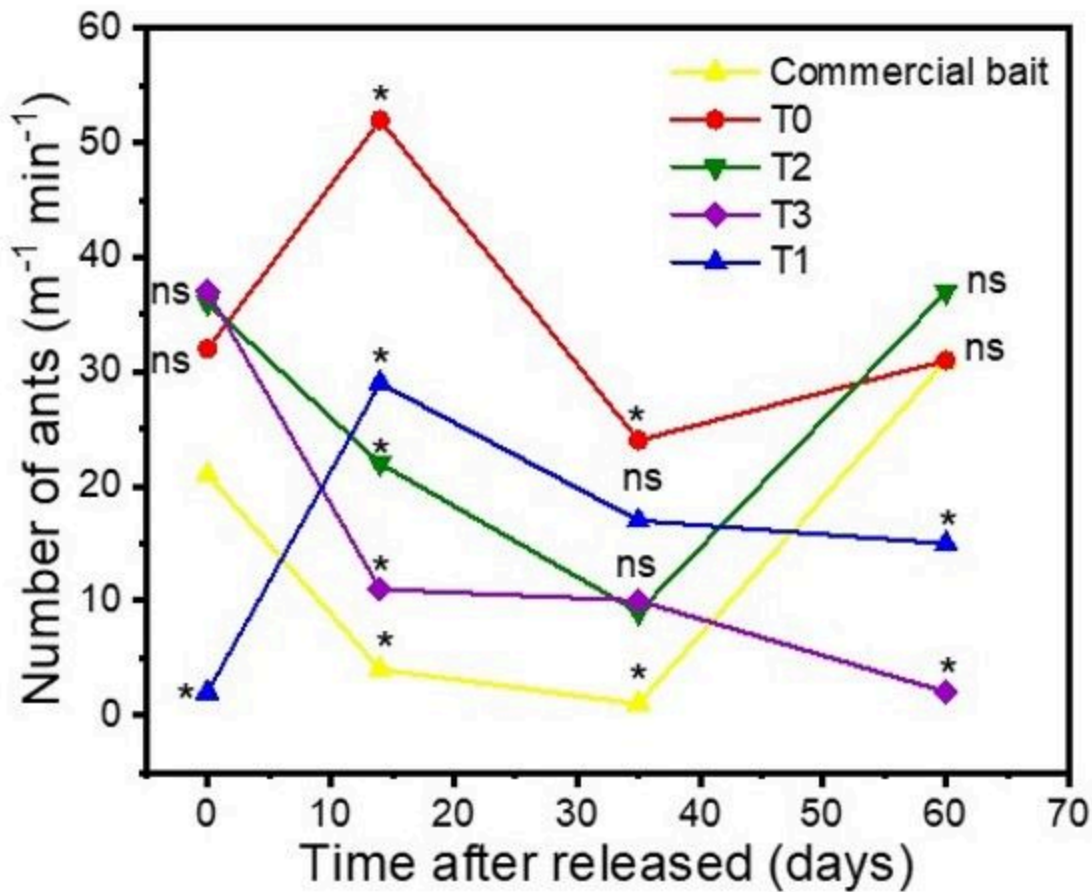


Figure 7

The flow of ants in the trails as a function of time after bait release in the different treatments.

Treatments - T0: control bait with 2.5 g of sodium tetraborate, T1: bait with 2.5 g of sodium tetraborate and Boveril[®], T2: bait 5.0 g of sodium tetraborate and Boveril[®], T3: bait with 2.5 g of sodium tetraborate and chlorpyrifos and commercial bait. The symbol (*) indicates a significant difference, ns= not significant (T test, P<0.05).